World-Model Value Depends on the Policy That Acts

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OUR POSITION

A better policy can gain more from identical world-model advice.

Evaluate predictive benefit jointly with the policy that acts.

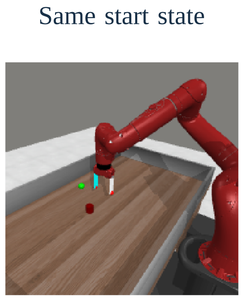
+11.50 mm

Change in advice gain
BC100 → BC300

01 Hold the advice fixed; change the policy that follows.

Task: push the red object to the green goal (Meta-World [1]). Frozen predictor: JEPA-WM [2]. BC = behavior cloning.

Action advice: execute the first five commands of a selected plan; the tested policy then controls the robot from new observations.



Red object; green goal

Same start state

Execute five commands
Steps 10–15

- BC300 baseline commands → Tested BC takes over
- WM-selected commands → Tested BC takes over

Policy resumes control
Same policy in both runs; steps 15–100

Measure final distance
Average over steps 91–100

- Baseline distance
- Distance with advice

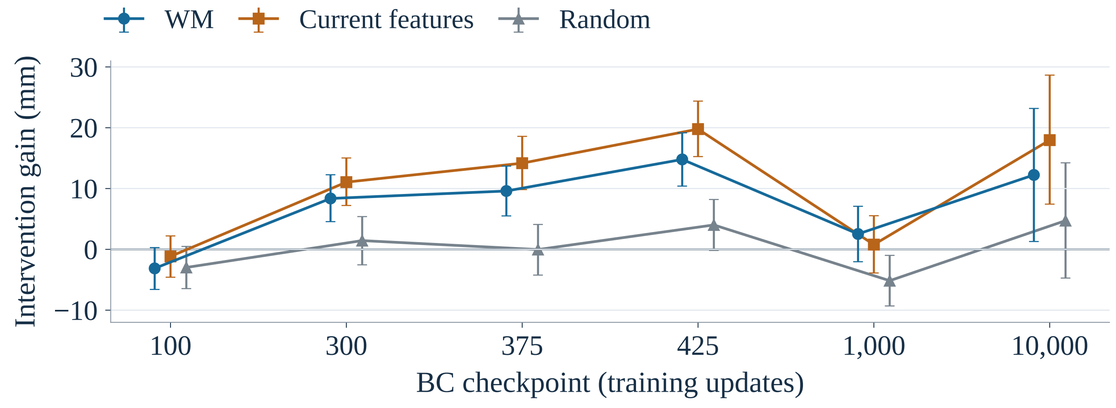
Gain = baseline − advice

Held fixed across policies: the state, observations, forecasts, candidate plans, baseline commands, and selected commands.

Selection: forecast outcomes for nine 20-step plans → learned outcome scorer → execute five commands. Control: score current features with its own validated threshold.

02 Better control, greater advice value

512 shared states; one five-command intervention.
BC100/300 denote 100/300 training updates.



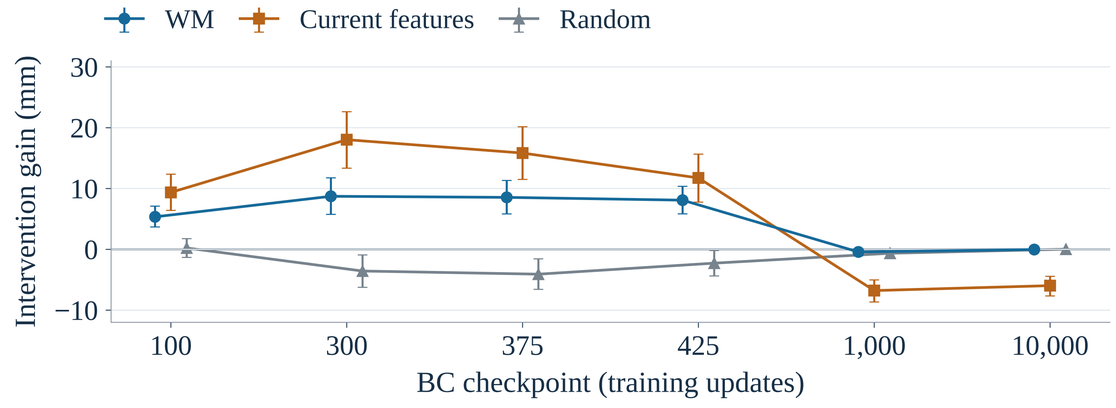
Final distance and gain (mm)

Policy	Baseline ↓	With advice ↓	Gain ↑
BC100	143.51	146.64	−3.13
BC300	121.08	112.71	**+8.37**

Baseline improves **22.43 mm**; advice gain rises **11.50 mm**.
Gain increase: paired 95% CI [6.80, 16.25] mm.

03 Separate benefit from predictive advantage

1,024 cases; repeated selection at visited states.
Baseline: BC alone throughout.



+8.73 vs. +18.04 mm

WM vs. current-feature scoring at BC300.

Current features give the larger gain.

Intervention benefit does not establish predictive advantage.
Shared encoder, candidate rule, capacity, and supervision.

Metric: gain = baseline − intervention distance, averaged over steps 91–100; positive is better. Bars: paired 95% intervals. Protocols differ in targets, thresholds, schedules, and baselines.
Fixed-advice evidence: BC100 → BC300 was fixed before fresh evaluation. Gain at BC300, gain increase, and baseline improvement have positive family-adjusted lower bounds (15 tests; α = .010).

04 Short-term gains can be reversed

Use actual simulated endpoints to select one of nine 20-step plans.
Execute all 20 actions, then resume BC1000. Baseline: unmodified 20-step plan.

20-step gain	→	Final-task gain
+7.79 mm		−3.68 mm
95% CI [7.15, 8.48]		95% CI [−6.15, −1.74]

Evaluate the outcome after policy control resumes.

Post hoc; 256 separate cases; BC inputs differ from the primary study.

05 Action–forecast matching matters

Exploratory; 512 reused states; five-step forecasts; BC300.
Scorers also receive BC300's five actions at each of eight probe states.

Matched WM	Shuffled WM	Current features
17.08 mm	5.48 mm	15.13 mm

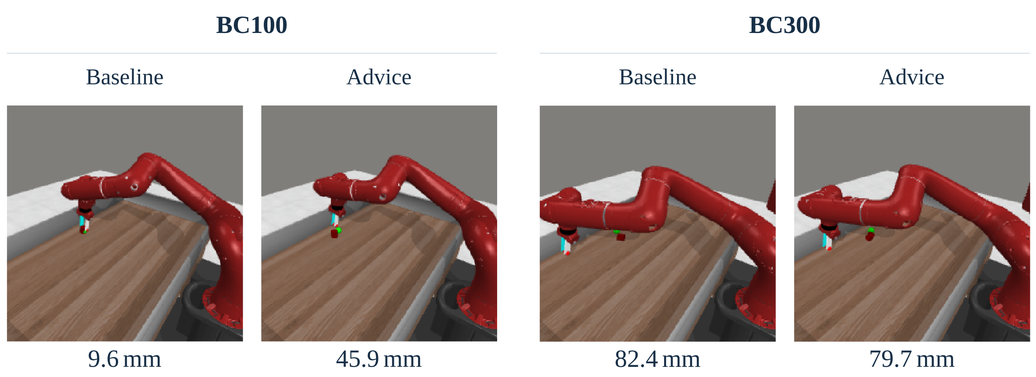
Matched − shuffled: **+11.60 mm** [7.37, 15.90].

Shuffling breaks action–forecast pairing. Gains shown above;
WM superiority to current features is unresolved (paired 95% CIs).

06 Robot diagnostics: identical advice, different consequences

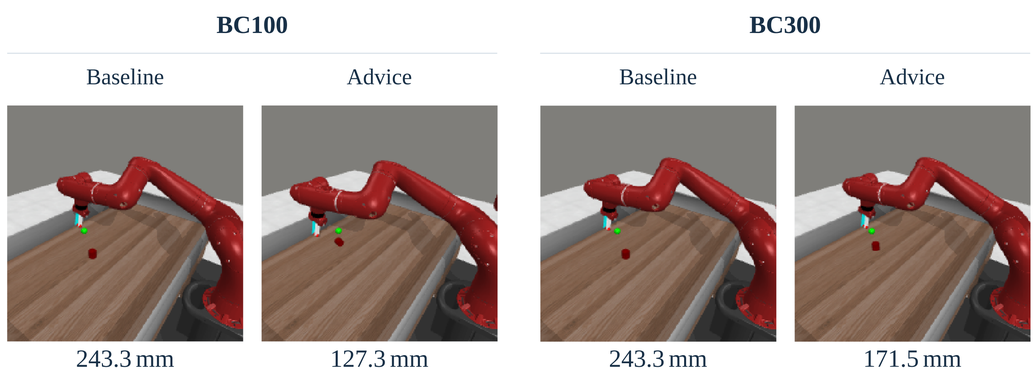
Separate exploratory examples; the scorer receives the same BC300 action description for both continuation policies.

(a) Greater gain through reduced harm



Gain: −36.3 → +2.6 mm (baseline worsens)

(b) Smaller benefit from the same advice



Gain: +116.0 → +71.8 mm

Actual simulated states at step 100; labels average steps 91–100. Cases chosen after evaluation near each sign group's median gain change.
Across 512 cases: 252 increased / 177 decreased / 83 unchanged gains. Increased gain alone need not imply better baseline control.

Evaluate the model–policy pair through final task outcomes.

Report baseline and gain separately; compare current-feature selection; account for inference cost.

One task/BC run; privileged observations; physics paused. BC300 candidate evaluation: WM 1.001 s vs. direct .057 s/episode.

[1] Yu et al., *Meta-World*, CoRL 2020. [2] Terver et al., *What Drives Success in Physical Planning with Joint-Embedding Predictive World Models?*, 2025.

QR: Paper